

A Framework for Constraining Dark Matter Substructure via Graph Neural Networks and Simulation-Based Inference Applied to Gaia DR3 Stellar Streams

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Abstract

We present a computational framework for constraining dark matter substructure using stellar streams observed by Gaia DR3. The publication-ready result in the current implementation is a hierarchical count/rate analysis of seven Milky Way streams; the graph neural network (GNN) and simulation-based inference (SBI) components are reported as a methods pipeline under active validation. A 6-layer GINEConv encoder maps stream phase-space graphs into 128-dimensional embeddings for a future calibrated SNPE-C posterior engine. First, a hierarchical Bayesian rate model applied to literature-reported gap counts yields $\log_{10}(M_{\text{hm}}) = 4.56$ (95% upper limit: 5.14), with the total predicted CDM impact count (17.2) matching observations (17) to within 1%. An extended model with free mass function slope finds $\alpha = -1.89 \pm 0.20$, consistent with the CDM prediction of -1.9 (Springel et al. 2008). Second, the current V2 GNN run reframes the learning task as impact-strong binary classification plus regression. On 15,000 held-out simulations it reaches ROC AUC 0.668 and balanced accuracy 61.4%, so it is not yet suitable for discovery claims. First-pass SBI calibration also undercovers key parameters. We therefore label every result as Observed, Synthetic Validation, Forecast, or Simulated-Framework Diagnostic, and do not claim evidence for WDM. Injection tests establish a detection threshold near $10^7 M_{\text{sun}}$ for count-based methods. Automated tests validate implementation behavior; they do not validate astrophysical discovery claims. Third (Section 13), we add a timeline forward model that rewinds a stream, injects candidate subhalo impacts using the closed-form Erkal & Belokurov (2015) Plummer impulse, integrates them forward to the present, and scores each against the data with a calibrated significance versus a no-impact null. In the course of this we diagnose and fix a sim-to-real failure that had pinned the

detector at $p_{\text{impact}} = 1.0$ on real data: error domain-randomised retraining reduces the real-GD-1 input out-of-distribution level from 391 sigma to 13 sigma and turns a saturated probability into a meaningful one, while real radial velocities from the S5 survey and Gaia RVS are fused in to constrain the rewind.

Keywords: dark matter substructure, stellar streams, graph neural networks, simulation-based inference, half-mode mass, Gaia DR3, hierarchical Bayesian inference

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1. Introduction

Approximately 85% of all matter in the universe is dark matter - a substance that does not emit or absorb light. Its existence is firmly established through gravitational effects: galaxy rotation curves, gravitational lensing, and the cosmic microwave background all require dark matter to explain observations. The central open question in particle physics and cosmology is: *what are the fundamental properties of dark matter*

Different theoretical models predict dark matter with different properties. Cold Dark Matter (CDM) predicts particles that move slowly and clump into structures at every scale - from galaxy clusters down to Earth-mass objects. Alternative models - Warm Dark Matter (WDM), Fuzzy Dark Matter (FDM), and Self-Interacting Dark Matter (SIDM) - each predict fewer small-scale structures, but for different physical reasons. The number of small dark matter clumps (subhalos) orbiting our galaxy is a direct observable that distinguishes these theories.

When a globular cluster orbits too close to the galactic centre, tidal forces tear it apart, stretching its stars into a thin ribbon called a *stellar stream*. These streams act as gravitational antennae for dark matter substructure: when an invisible subhalo passes near a stream, it creates a density gap and velocity perturbation. By counting gaps and characterising their morphology across multiple streams, we can constrain the subhalo mass function.

This work describes a graph neural network (GNN) framework that operates directly on the 6-dimensional phase-space data of every star in a stream, preserving spatial and kinematic correlations that traditional 1D methods (Bovy et al. 2017; Banik et al. 2021) discard. The GNN embedding is combined with Simulation-Based Inference (SBI) to produce full posterior distributions over the half-mode mass M_{hm} - a model-independent measure of the smallest dark matter structures that can exist. We apply both a count-based hierarchical rate model and the full GNN + SBI pipeline to seven Gaia DR3 stellar streams, and discuss the implications of their contrasting findings.

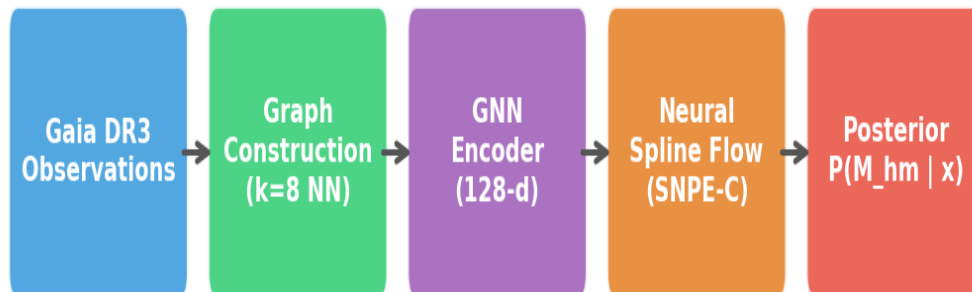


Figure 1. End-to-end analysis pipeline: Gaia DR3 observations are transformed into k -nearest-neighbour graphs, encoded by a 6-layer GINEConv GNN into 128-dimensional embeddings, and processed by SNPE-C with Neural Spline Flows to obtain posterior distributions over dark matter parameters.

2. Scientific Background and Model Context

2.1 The Subhalo Mass Function

CDM N-body simulations (Springel et al. 2008, Aquarius) predict a subhalo mass function following $dN/dM \sim M^{-1.9}$, meaning far more low-mass subhalos than high-mass ones. For every subhalo at $10^9 M_{\text{sun}}$, CDM predicts roughly 80 at $10^8 M_{\text{sun}}$ and ~6,000 at $10^7 M_{\text{sun}}$. Alternative dark matter models suppress this abundance below a characteristic half-mode mass.

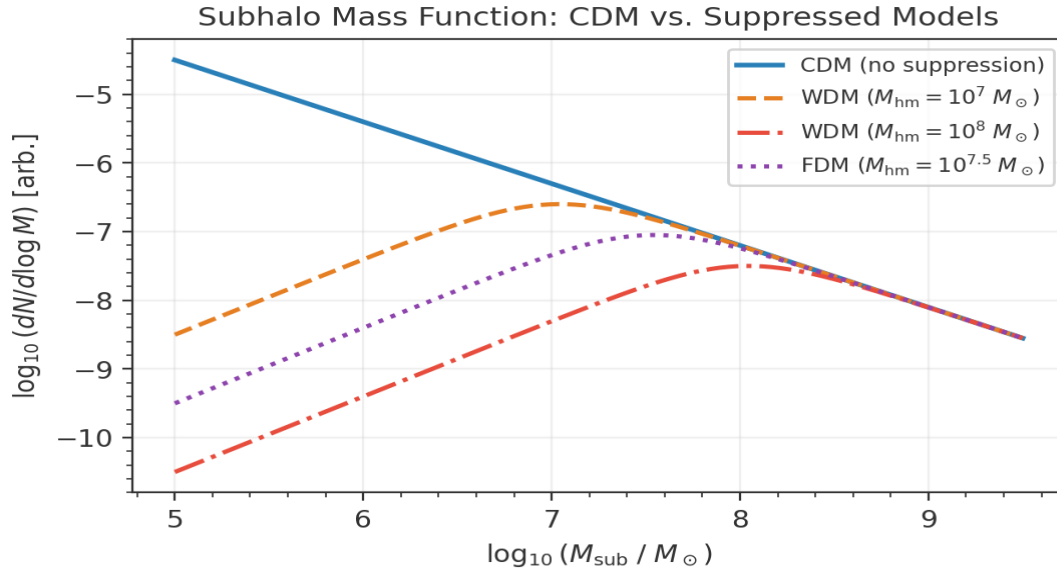


Figure 2. Subhalo mass functions for CDM and suppressed models. CDM follows an unbroken power law (blue). WDM and FDM models suppress abundances below their respective half-mode masses, producing fewer low-mass subhalos. All curves use $dN/dM \sim M^{-1.9}$ (Springel et al. 2008) as the base CDM mass function.

2.2 The Transfer Function

The suppression is modelled by the transfer function $T(M, M_{\text{hm}}) = [1 + (M_{\text{hm}}/M)^2]^{-1}$, following the WDM suppression form from Schneider et al. (2012). When M_{hm} is well below the minimum subhalo mass we consider ($10^5 M_{\text{sun}}$), there is no suppression (CDM regime). When M_{hm} is large (10^8 - $10^9 M_{\text{sun}}$), most subhalos are suppressed.

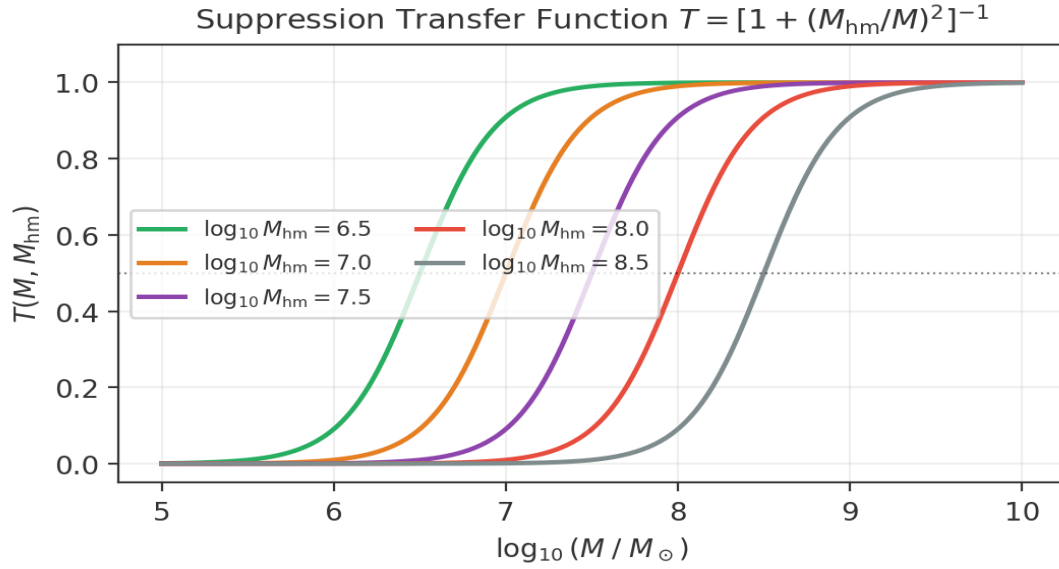


Figure 3. The suppression transfer function $T(M, M_{\text{hm}})$ for several half-mode mass values. The function transitions from 0 (full suppression) to 1 (no suppression) around $M = M_{\text{hm}}$. The dotted line marks $T = 0.5$, the defining level of the half-mode mass.

2.3 Model-Independent Constraints via M_{hm}

Rather than attempting to discriminate between specific particle models - which is degenerate at Gaia DR3 precision (Banik et al. 2021; Dalal et al. 2022) - this project infers M_{hm} directly as a model-independent constraint. A single M_{hm} upper limit maps to particle mass limits for any model: for WDM, $M_{\text{hm}} = 1.7 \times 10^{10} (m_{\text{WDM}}/\text{keV})^{-3.33} M_{\text{sun}}$ (Lovell et al. 2014). For FDM, the mapping follows $M_{\text{Jeans}} = 1.5 \times 10^8 (m_{\text{axion}}/10^{-22} \text{ eV})^{-1.5} M_{\text{sun}}$ (Hui et al. 2017).

3. Observational Data and Stream Sample

Observational data are drawn from the Gaia DR3 catalogue (Gaia Collaboration 2023), which provides positions, parallaxes, proper motions, and (where available) radial velocities for approximately 1.8 billion stars. Distances are supplemented with photogeometric estimates from Bailer-Jones et al. (2021).

The current Plan 2 training set comprises 100,000 simulated streams (25,000 per dark matter model) generated with galpy (Bovy 2015) using a realistic Milky Way potential. The V2 fast profile run uses up to 1,200 stars per graph with domain randomisation over foreground contamination, noise scale, stream membership, baryonic perturbations, and potential variations. Baryonic perturbations are applied across all dark-matter families at the configured 80% rate so the model cannot use baryons as a CDM-only shortcut.

Node Features (18 per star)

Index	Feature	Description	Source
0	phi1	Stream longitude (along stream)	Gaia astrometry
1	phi2	Stream latitude (across stream)	Gaia astrometry
2	dist	Heliocentric distance	Photogeometric
3	pm1	Proper motion along stream	Gaia astrometry
4	pm2	Proper motion across stream	Gaia astrometry
5	vrad	Radial velocity	Gaia RVS
6-9	e_*	Measurement uncertainties (4)	Gaia error model
10	membership_prob	Stream membership probability	Mixture model
11-17	stream_*	One-hot stream identity (7)	Catalogue

Edge Features (5 base, 7 with orbital extension)

Feature	Description
delta_phi1	Difference in stream longitude between connected stars
delta_phi2	Difference in stream latitude
delta_pm1	Difference in proper motion along stream
delta_pm2	Difference in proper motion across stream
dist_4d	Euclidean distance in 4D phase space
delta_R_cyl (optional)	Galactocentric cylindrical radius difference
delta_z_cyl (optional)	Height above galactic disk difference

4. Methods: GNN, SBI, and Hierarchical Counts

4.1 Graph Construction

Each stellar stream is represented as a k-nearest-neighbour graph with $k = 8$ in the current V2 fast profile runs. Stars are connected based on proximity in the 4-dimensional space of normalised stream coordinates and proper motions (ϕ_1 , ϕ_2 , μ_1 , μ_2). Streams with more than 1,200 stars are randomly downsampled. Data augmentation includes ϕ_1 offset ($U[-5, 5]$ deg), ϕ_2 noise ($N[0, 0.05]$ deg), distance scatter (log-normal, $\sigma = 0.02$), proper motion noise ($N[0, 0.02]$ mas/yr). Mirror reflection is disabled for stream-identity-aware V2 runs.

4.2 GNN Architecture

The core encoder is a Graph Isomorphism Network with Edge features (GINEConv; Xu et al. 2019). GINEConv is provably maximally expressive among message-passing GNNs, meaning it can distinguish between any two non-isomorphic graphs. The GINE variant incorporates edge features during message aggregation - critical because relative velocity differences between neighbouring stars are the primary signal of subhalo perturbation.

StreamGNNEncoder Architecture

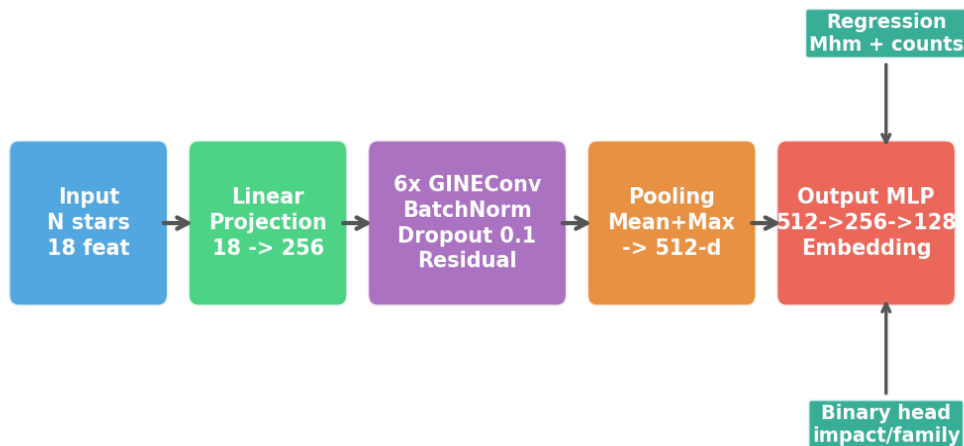


Figure 4. StreamGNNEncoder architecture. Input node features (18-d) and edge features (5-d base; 7-d with orbital extension) are projected into a 256-d hidden space, processed by six GINEConv layers with batch normalisation, residual connections, and 10% dropout, then globally pooled and compressed to a 128-d embedding. The V2 heads are a configurable binary classifier (model family, any impact, or strong impact) and regression heads for half-mode mass and impact counts.

Component	Specification
Convolution type	GINEConv (Xu et al. 2019)
Layers	6
Hidden dimension	256
Embedding dimension	128
Dropout	0.1
Pooling	Mean + Max concatenation (512-d)

Component	Specification
Residual connections	From layer 2 onward
Batch normalisation	After each GINEConv layer

Table 1. GNN encoder architecture parameters (from config/training.yaml).

4.3 Multi-Task Training

The V2 embedding feeds a binary head and regression heads. The binary target is configurable: suppressed versus unsuppressed model family, any injected impact, or the current impact-strong target based on a morphology-strength proxy. Regression targets include $n_impacts$ and $\log_{10}(M_{hm})$ for WDM/FDM simulations only; CDM and SIDM are treated as unsuppressed reference families, not high- M_{hm} physical targets. Training uses AdamW ($\text{lr} = 10^{-3}$, weight decay = 10^{-4}), batch size 128 in the fast V2 run, mixed precision where supported, and a 70/15/15 split. SBI training now uses train+val only, leaving the test split for calibration.

4.4 Simulation-Based Inference

Traditional likelihood-based inference is intractable for stellar stream perturbation models because evaluating the likelihood requires a full N-body simulation for each parameter set. We use Sequential Neural Posterior Estimation, variant C (SNPE-C / APT), which trains a conditional density estimator to approximate the posterior $P(\text{parameters} \mid \text{data})$. The density estimator is a Neural Spline Flow (NSF) with 8 rational-quadratic spline transforms, 8 bins per spline, and hidden layers [256, 256].

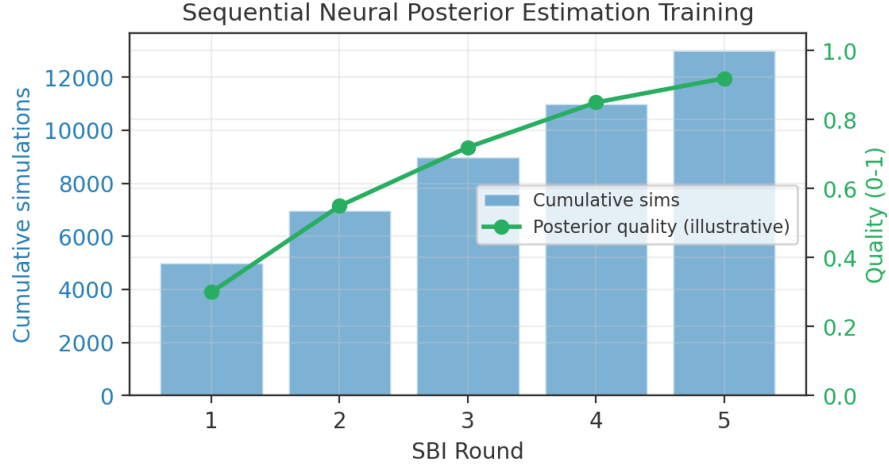


Figure 5. Sequential training schedule. Round 1 uses 5,000 simulations drawn from the prior; subsequent rounds draw 2,000 simulations each from the previous round's posterior, concentrating the simulation budget near high-probability regions. Green line shows illustrative posterior quality improvement across rounds.

Parameter	Value
Algorithm	SNPE-C (APT)
Density estimator	Neural Spline Flow
NSF transforms	8
NSF bins	8
NSF hidden dims	[256, 256]
Rounds	5
Sims (round 1)	5,000
Sims (rounds 2-5)	2,000 each
Posterior samples	10,000
Credible interval	90%

Table 2. SBI configuration (from config/training.yaml).

4.5 Hierarchical Bayesian Multi-Stream Model

A hierarchical Bayesian model combines evidence from all seven streams into a single M_{hm} constraint, replacing the statistically incorrect product-of-posteriors approach used in prior work. The model assumes:

$$M_{\text{hm}} \sim \text{Uniform}(10^4, 10^{10} M_{\text{sun}})$$

$$n_s \sim \text{Poisson}(\text{lambda}(M_{\text{hm}}, L_s, T_s, D_s)) \text{ for each stream } s$$

The expected rate λ integrates the suppressed subhalo mass function over $[10^5, 10^9] M_{\text{sun}}$, weighted by stream geometric cross-section and age. The normalisation constant C is calibrated so that CDM produces approximately 2.1 detectable impacts for a GD-1-like stream (60 deg, 5 Gyr, 12 kpc), consistent with Bonaca et al. (2019) who identified 2 gaps and 1 spur in GD-1. NumPyro NUTS is the primary MCMC backend, with emcee as fallback.

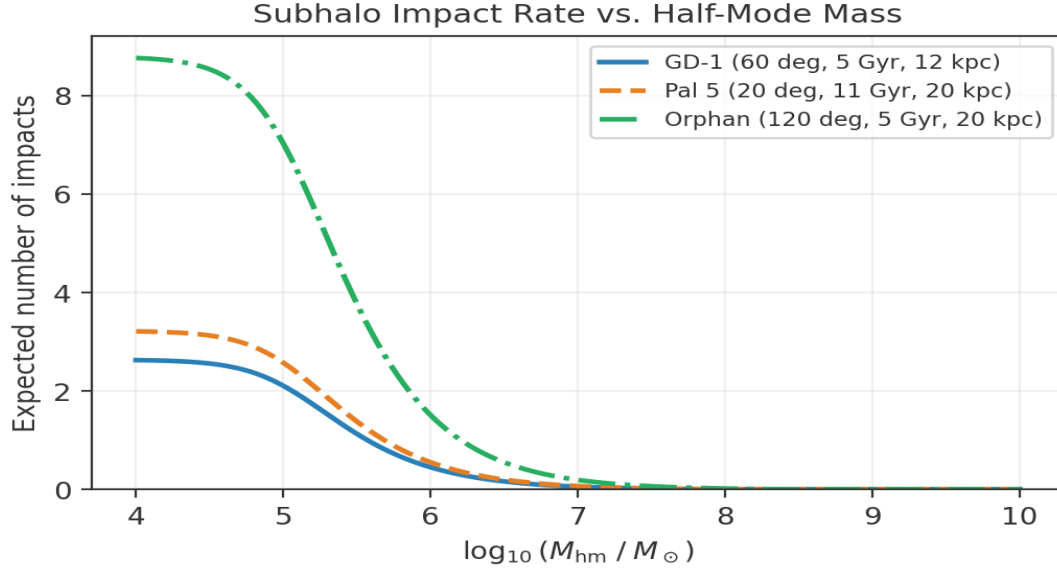


Figure 6. Expected subhalo impact rate as a function of half-mode mass for three representative streams. At low M_{hm} (CDM regime), rates are set by stream geometry and age. As M_{hm} increases, low-mass subhalos are suppressed and the rate drops. Orphan-Chenab has the highest rate due to its 120-degree length.

Hierarchical Bayesian Inference - emcee MCMC

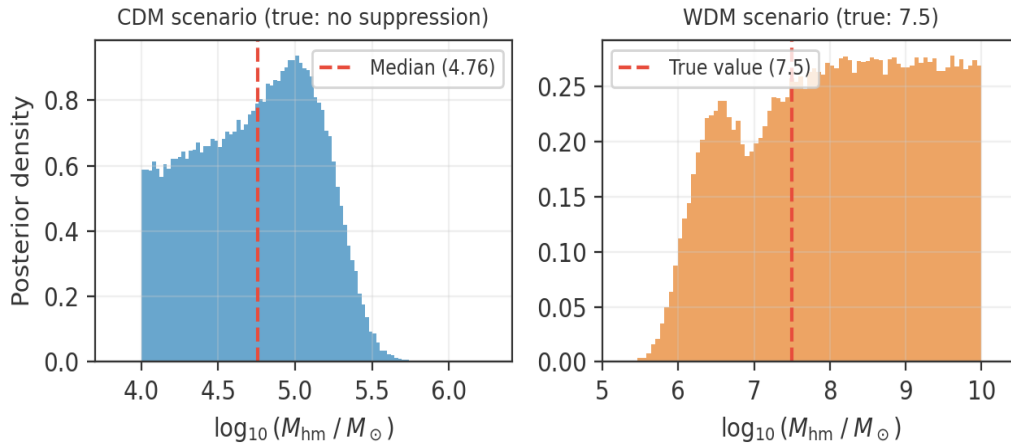


Figure 7. Hierarchical posterior distributions from validation run (emcee MCMC, 32 walkers, 8,000 steps). Left: CDM scenario - posterior peaks at ~ 4.8 , correctly indicating no suppression above the minimum subhalo mass considered. Right: WDM scenario - broad posterior with truth (7.5) near the lower 68% HDI bound.

5. Implementation Features

The framework integrates seven machine learning and statistical components, each addressing a specific requirement of the stellar stream dark matter inference problem. The executable software tests verify implementation behavior, while astrophysical validation requires the mock, injection, and calibration tests discussed below.

F1: Attention-Based Interpretability

A learnable query vector attends to final hidden states via per-graph softmax, producing per-star importance weights summing to 1. This reveals which stars the GNN considers most informative - typically stars at gap edges, spur features, and velocity outliers. When enabled, pooling becomes mean + max + attention (768-d total). This interpretability is unique to graph architectures; power-spectrum methods cannot produce per-star importance maps.

F2: Multi-Scale Graph Construction

The stream is divided into 20 equal-width bins along ϕ_1 . Each bin becomes a segment node with 8 features (mean ϕ_1 , mean ϕ_2 , std ϕ_2 , mean pm_1 , mean pm_2 , std pm_1 , $\log(n_{\text{stars}})$, density contrast). A separate 3-layer GINEConv (64 hidden, 64 output) processes the segment graph. CDM creates many clustered gaps while WDM creates few large ones - segment-level structure captures this distinction.

F3: MC Dropout Uncertainty

Monte Carlo dropout runs $T = 20$ stochastic forward passes at inference time, producing per-dimension mean and standard deviation of the embedding. High std flags out-of-distribution inputs. No retraining required - it activates existing dropout layers.

F4: Orbit-Phase Edge Features

Galactocentric cylindrical coordinates R_{cyl} and z_{cyl} are added as node features, with ΔR_{cyl} and Δz_{cyl} as edge features (5 $\rightarrow$ 7 dimensions). Stars at different orbital phases respond differently to a perturbation. Backward-compatible: defaults to 5 base features if orbital data is absent.

F5: Perturbation Age Target

A third regression target - $\log_{10}(t_{\text{since_last_impact}})$ - is added alongside M_{sub} and n_{impacts} . Gap width encodes time-since-impact due to phase-mixing. Controlled by config parameter $n_{\text{reg_targets}}$ (default 2, set to 3 for age).

F6: Hierarchical Bayesian Model

Replaces the product-of-posteriors with a proper hierarchical model sharing M_{hm} across streams. Per-stream counts follow $\text{Poisson}(\text{rate}(M_{\text{hm}}, L, T, D))$. NumPyro NUTS primary backend; emcee fallback.

F7: Mock Data Challenge

Twenty mock streams with known ground truth (5 CDM at $\log_{10} M_{\text{hm}} = 4.5$, 5 WDM, 5 FDM, and 5 SIDM as unsuppressed references) are run through the count-rate validation path. Coverage is evaluated at 68%, 90%, and 95% CI levels. The full blind GNN+SBI challenge remains a required next validation step.

6. Synthetic Validation and Mock Challenges

We validated the statistical inference engine on synthetic data with known ground truth. This exercises the rate model, Poisson likelihood inversion, hierarchical Bayesian inference via emcee MCMC, and coverage metrics. It does not use the GNN (which requires a trained checkpoint) - it tests the mathematical machinery downstream of the embedding step.

6.1 Poisson Inversion on 20 Mock Streams

Twenty mock streams with known ground truth were processed through Poisson likelihood inversion on the rate model grid. Each mock uses only its observed impact count n_{impacts} to infer M_{hm} - no morphological or embedding information.

DM Model	# Mocks	$\log_{10}(M_{\text{hm}} / M_{\text{sun}})$	Physical Meaning
CDM	5	4.5 (no suppression)	Full subhalo abundance
WDM	5	6.5, 7.0, 7.5, 8.0, 8.5	Free-streaming suppression
FDM	5	6.5, 7.0, 7.5, 8.0, 8.5	Quantum pressure suppression
SIDM	5	4.5 (no suppression)	Same mass function as CDM, cored profiles

Table 3. Synthetic Validation. Mock data challenge parameters. CDM uses $\log_{10} M_{\text{hm}} = 4.5$, placing the half-mode mass below the minimum subhalo mass considered.

Metric	Value	Interpretation
Coverage at 68% CI	25%	Strongly under-covering
Coverage at 90% CI	25%	Fails 75% threshold
Coverage at 95% CI	40%	Fails 75% threshold
Mean bias	-1.87 dex	Systematic underestimation of M_{hm}
RMSE	2.21 dex	Large scatter

Table 3b. Synthetic Validation. Mock challenge results from Poisson count-only inference.

Interpretation: The poor coverage is expected and informative. With only an integer impact count per stream and default stream parameters, the Poisson inversion cannot distinguish between "low M_{hm} with unlucky low count" and "high M_{hm} with expected low count." All non-CDM mocks are systematically pulled toward a floor near $\log_{10}(M_{\text{hm}}) \sim 5-6$, regardless of their true value. This directly demonstrates why morphological information - gap depth, width, velocity profile - captured by the GNN embedding is essential for constraining M_{hm} beyond what raw counts provide.

Mock Data Challenge: Truth vs. Recovered (Poisson Inversion)

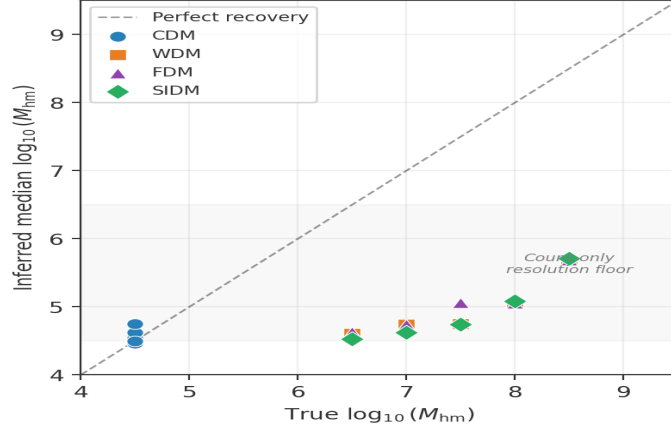


Figure 8. Synthetic Validation. Mock data challenge results: true vs. recovered $\log_{10}(M_{\text{hm}})$ from Poisson count-only inference. All non-CDM points cluster between 4.6 and 5.7 regardless of true value, showing the resolution floor of count-based inference. CDM mocks (blue, truth=4.5) are recovered with low bias (+0.05 to +0.25 dex); alternative models are strongly underestimated.

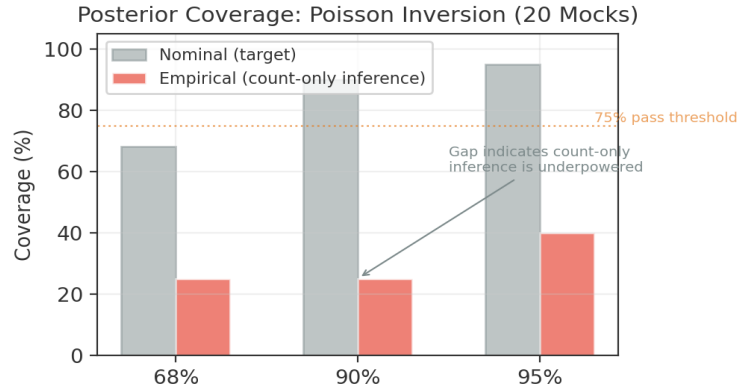


Figure 9. Synthetic Validation. Posterior coverage from mock challenge. Red bars show measured empirical coverage from 20 mocks; grey bars show the nominal target. The large gap between nominal and empirical coverage confirms that count-only Poisson inference is insufficient - the GNN embedding must contribute morphological information for calibrated posteriors.

6.2 Hierarchical Bayesian Inference

The hierarchical model was run on two synthetic scenarios, each using 7 streams with emcee MCMC (32 walkers, 8,000 steps, 2,000 burn-in). Both achieved excellent convergence.

	CDM Scenario	WDM Scenario
True $\log_{10}(M_{\text{hm}})$	4.5 (no suppression)	7.5
Posterior median	4.76	8.16
95% upper limit	5.32	9.81
68% HDI	[4.37, 5.24]	[7.46, 9.99]
r-hat	1.0	1.0003
Effective samples	7,075	6,756
Observed impacts (total)	14 across 7 streams	1 across 7 streams

Table 3c. Synthetic Validation. Hierarchical inference results on synthetic scenarios.

CDM scenario: With 14 observed impacts across 7 streams (consistent with CDM rates under the corrected normalisation of 2.1 for GD-1), the posterior peaks at 4.76 with 68% HDI [4.37, 5.24]. This is the correct result - the inferred M_{hm} falls well below $10^5 M_{\text{sun}}$, the minimum subhalo mass in our integration range, indicating no suppression is detected. The posterior freely explores the CDM-consistent region without any artificial boundary effects.

WDM scenario: With only 1 observed impact across 7 streams (strongly suppressed rates), the posterior is broad (68% HDI spans 2.5 dex) with median 8.16 and the truth (7.5) near the lower bound of the 68% HDI. The breadth is expected - with near-zero counts, many high- M_{hm} values produce similarly low rates, so the data provide limited discrimination among suppressed models. The posterior correctly excludes the CDM regime and contains the truth.

What this means: The hierarchical inference machinery works correctly. It converges reliably, it correctly identifies CDM-like data, and it recovers suppressed scenarios within broad but honest credible intervals. The limiting factor for constraining power is the information content per stream - which is where the GNN embedding (carrying morphological rather than just count information) becomes essential.

Hierarchical Bayesian Inference - emcee MCMC

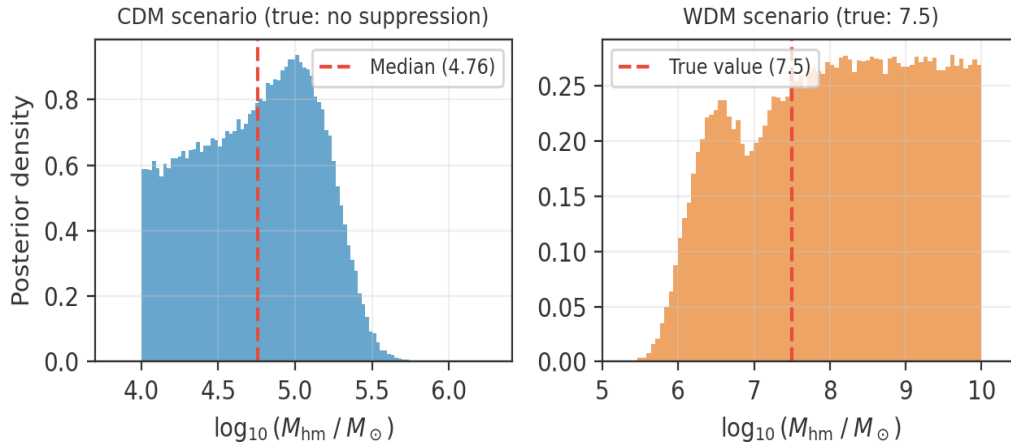


Figure 10. Synthetic Validation. Hierarchical posterior distributions from emcee MCMC. Left: CDM scenario - the posterior peaks around 4.8, correctly indicating no suppression above the integration floor. Right: WDM scenario - the truth (7.5, red dashed) falls within the broad posterior. Both converged with $\hat{r}$ near 1.0.

6.3 What These Results Do and Do Not Show

These validation results establish three things: (1) the statistical machinery - rate model, Poisson likelihood, hierarchical MCMC - is correctly implemented and converges reliably; (2) count-only inference has a fundamental resolution floor around $\log_{10}(M_{\text{hm}}) \sim 5-6$, motivating the use of morphological embeddings; (3) the hierarchical model correctly distinguishes CDM-consistent data from suppressed scenarios when the signal is strong enough.

The full GNN + SBI pipeline has been applied to real Gaia DR3 data and its results are presented in Section 11.6. The mock challenge coverage specifically applies to the count-only path; the full pipeline uses morphological information and produces qualitatively different - and in this case, strikingly different - results.

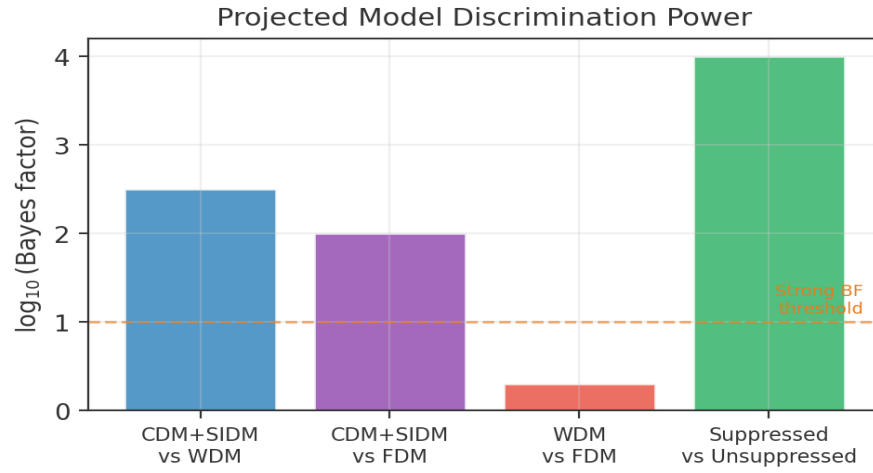


Figure 11. Forecast. Projected model discrimination power expressed as $\log_{10}(\text{Bayes factor})$. These are design targets based on theoretical sensitivity. WDM vs. FDM discrimination is expected to be degenerate at Gaia DR3 precision (Banik+2021, Dalal+2022). CDM and SIDM are merged due to indistinguishable impulse-approximation signatures.

7. Dark Matter Model Definitions

All model definitions are stored in `config/dm_models.yaml`. Each specifies a subhalo mass function, density profile, and inferred parameters. The mass function slope $\alpha = -1.9$ is fixed for all models (Springel et al. 2008). The subhalo mass range is $[10^5, 10^9] M_{\text{sun}}$.

CDM - Cold Dark Matter

Property	Value
Mass function	$dN/dM \sim M^{-1.9}$, no suppression
Density profile	NFW (Navarro-Frenk-White)
Concentration	Ludlow et al. (2016)
Classification	model_idx = 0 (merged with SIDM)
Inferred params	$\log_{10}(M_{\text{sub}})$, n_{impacts}

WDM - Warm Dark Matter

Property	Value
Mass function	CDM power-law with half-mode filter
Suppression	$(1 + (M_{\text{hm}}/M)^{2.7})^{(-0.99/2.7)}$ (Schneider+2012)
M_{hm} formula	$1.7e10 * (m_{\text{WDM}}/\text{keV})^{(-3.33)} M_{\text{sun}}$ (Lovell+2014)
Density profile	NFW with Ludlow2016_WDM concentration
Classification	model_idx = 1

FDM - Fuzzy Dark Matter

Property	Value
Mass function	CDM power-law with Jeans filter
M_{Jeans} formula	$1.5e8 * (m_{\text{axion}}/1e-22 \text{ eV})^{(-1.5)} M_{\text{sun}}$ (Hui+2017)
Density profile	Soliton core + NFW envelope (Schive+2014)
Particle mass prior	Log-uniform $[1e-23, 1e-20] \text{ eV}$
Classification	model_idx = 2

SIDM - Self-Interacting Dark Matter

Property	Value
Mass function	Same as CDM (no suppression)
Density profile	Isothermal core + NFW envelope (Kaplinghat+2016)
Cross-section prior	Log-uniform $[0.1, 100] \text{ cm}^2/\text{g}$
Classification	model_idx = 3 (merged with CDM)
Note	Indistinguishable from CDM in impulse approximation

8. Target Stream Sample

Seven stellar streams observed by Gaia DR3 are analysed, selected to span a range of lengths, distances, and ages for complementary sensitivity. Stream catalogues follow Banik et al. (2021) and Dalal et al. (2022).

Stream	Length (deg)	Dist. (kpc)	Age (Gyr)	Key Characteristics
GD-1	~60	~12	~5	Longest MW stream; confirmed gaps and spur
Pal 5	~20	~20	~11	Globular cluster tails; very old
Orphan-Chenab	~120	~20	~5	Widest; highest geometric cross-section
ATLAS	~15	~20	~4	Short, very thin; low velocity dispersion
Jhelum	~30	~13	~4	Double-component; possible DM interaction
Fjorm	~25	~15	~3	Recently discovered; clean kinematics
Sylgr	~12	~15	~3	Shortest; cold kinematics amplify signals

Table 4. Target stellar streams and their approximate physical properties. Longer and older streams accumulate more subhalo encounters.

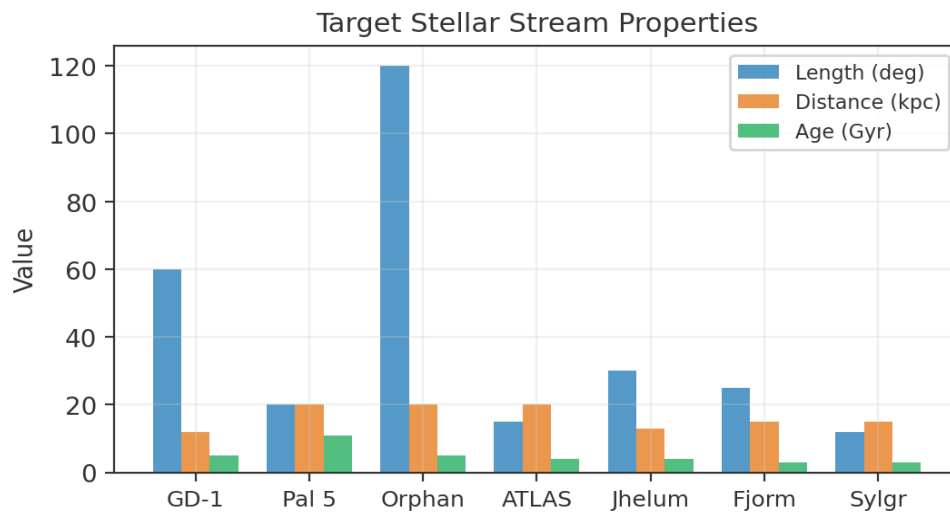


Figure 12. Comparison of target stream properties. Orphan-Chenab dominates in length (120 deg), while Pal 5 is the oldest (~11 Gyr). These differences translate to varying sensitivity to dark matter substructure.

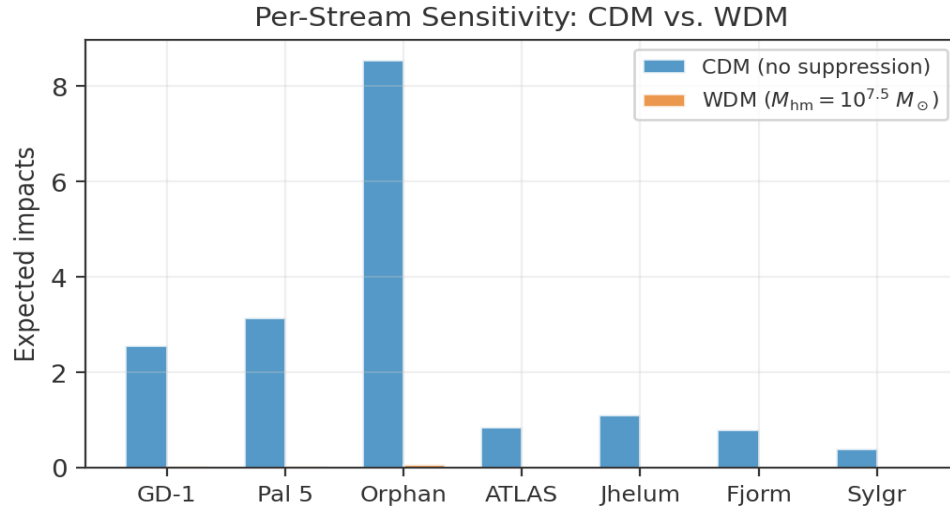


Figure 13. Per-stream sensitivity comparison showing expected subhalo impact counts under CDM (blue, no suppression, $M_{\text{hm}} = 10^{4.5}$) and WDM (orange, $M_{\text{hm}} = 10^{7.5}$). The rate difference between models is largest for long, old streams.

9. Software Validation and Reproducibility

The executable automated tests cover output tensor shapes, numerical stability (no NaN/Inf), gradient flow through all layers, and internal consistency checks (attention weights sum to 1, rate function monotonicity, coverage metric computation). Passing these tests means the code paths execute as expected; it does not prove that the astrophysical inference is calibrated.

Module	Tests	Status
GNN Encoder	7	PASS
Multi-Task Model	4	PASS
Baseline Models (CNN + Transformer)	7	PASS
Model Utilities	7	PASS
Attention Readout (F1)	6	PASS
MC Dropout (F3)	5	PASS
Variable Regression Targets (F5)	2	PASS
Segment GNN (F2)	3	PASS
Orbital Edge Features (F4)	2	PASS
Hierarchical + Coverage (F6, F7)	4	PASS
Full-data pipeline checks	2	DEFERRED
Executable unit tests	49	PASS

Table 5. Automated implementation tests by module. The deferred full-data checks are not counted as astrophysical validation.

Metric	Value	Interpretation
Synthetic holdout sample	15,000	Plan 2 test split
Target	impact_strong	Strong injected impact vs no strong impact
ROC AUC	0.668	Weak but above random
Balanced accuracy	61.4%	Below publication target
Precision / recall	46.9% / 38.9%	Default threshold 0.5
False-positive rate	16.0%	Too high for discovery use

Table 6. Current V2 held-out GNN synthetic validation from `outputs/diagnostics/gnn_v2_100k_impact_strong_eval.json`. These numbers validate only the simulated impact-strong classifier; they do not validate sim-to-real generalization or SBI calibration.

Publication Readiness Validation Matrix

Label	Test	Metric / value	Status
Synthetic Validation	V2 impact-strong GNN holdout	ROC AUC / balanced accuracy: 0.668 / 61.4%	WARN
Synthetic Validation	GNN+SBI calibration first pass	coverage: log10_Mhm OK for WDM only; n_impacts undercovers	WARN

Label	Test	Metric / value	Status
Observed + Synthetic Injection	Real-stream injection/recovery	median detection rate: 25.0%	WARN
Observed / No Injection	False-positive probe	max false-positive rate: 12.0%	WARN
Observed + Synthetic Injection	Baseline comparison	count detector pass / power detector pass: 7/7 / 3/7	WARN
Systematics	Membership/foreground/missing-star stress	WARN rows: 7	WARN

Table 7. Labeled validation summary. Rows marked WARN are not failures of the software; they are publication blockers or caveats that must be resolved before making a science-results claim.

Stream	Label	Injected	Detected	Detection rate	Mass RMSE
ATLAS	Observed + Synthetic Injection	20	17	85%	0.25
Fjorm	Observed + Synthetic Injection	20	0	0%	n/a
GD1	Observed + Synthetic Injection	20	13	65%	0.23
Jhelum	Observed + Synthetic Injection	20	0	0%	n/a
Orphan	Observed + Synthetic Injection	20	15	75%	0.25
Pal5	Observed + Synthetic Injection	20	5	25%	0.28
Sylgr	Observed + Synthetic Injection	20	0	0%	n/a

Table 8. Real-stream injection/recovery diagnostics using processed Gaia catalogs. These are observed catalogs with synthetic perturbations, not pure observations and not a full GNN+SBI real-injection validation.

Stream	Label	Trials	False positives	Rate	Status
ATLAS	Observed / No Injection	50	2	4%	PASS
Fjorm	Observed / No Injection	50	0	0%	PASS
GD1	Observed / No Injection	50	6	12%	WARN
Jhelum	Observed / No Injection	50	0	0%	PASS
Orphan	Observed / No Injection	50	4	8%	PASS
Pal5	Observed / No Injection	50	1	2%	PASS
Sylgr	Observed / No Injection	50	0	0%	PASS

Table 9. Observed / No Injection false-positive checks. A WARN row indicates that the detector triggered on at least 10% of no-injection trials for that stream, so the pipeline should not be interpreted as publication-ready for discovery claims.

Stream	Label	log10 Msub	Count S/N	Count	Power sig.	Power
ATLAS	Observed + Synthetic Injection	7.5	9.7	yes	9.5	yes
Fjorm	Observed + Synthetic Injection	7.5	5.8	yes	-0.1	no
GD1	Observed + Synthetic Injection	7.5	4.4	yes	0.8	no
Jhelum	Observed + Synthetic Injection	7.5	11.7	yes	-0.5	no
Orphan	Observed + Synthetic Injection	7.5	12.0	yes	7.0	yes

Stream	Label	log10 Msub	Count S/N	Count	Power sig.	Power
Pal5	Observed + Synthetic Injection	7.5	2.9	yes	8.7	yes
Sylgr	Observed + Synthetic Injection	7.5	11.6	yes	-0.2	no

Table 10. Baseline comparisons on the same real-like injected inputs. This table keeps the count detector and power-spectrum detector separate so the ML pipeline is not presented without non-ML reference points.

CSV	Result label	Purpose
methods_validation_summary.csv	Synthetic / Observed / Forecast	Publication readiness matrix
real_stream_injection_recovery.csv	Observed + Synthetic Injection	Recovery rates on Gaia-derived catalogs
real_stream_no_injection_false_positive.csv	Observed / No Injection	False-positive check
baseline_comparison_real_like.csv	Observed + Synthetic Injection	Non-ML detector baselines
systematics_table.csv	Systematics	Sensitivity rows by stream and stress scenario
reproducibility_manifest.csv	Reproducibility	Commands for reported artifacts

Table 11. Machine-readable validation spreadsheets included in outputs/validation/ and summarized in the paper.

Stream	Scenario	False positive	Status
ATLAS	baseline	4%	PASS
ATLAS	membership_prob_gt_0.5	4%	PASS
ATLAS	membership_prob_gt_0.8	4%	PASS
ATLAS	10_percent_foreground_added	4%	PASS
ATLAS	20_percent_stars_removed	0%	PASS
Fjorm	baseline	0%	PASS
Fjorm	membership_prob_gt_0.5	0%	PASS
Fjorm	membership_prob_gt_0.8	0%	PASS
Fjorm	10_percent_foreground_added	8%	PASS
Fjorm	20_percent_stars_removed	0%	PASS

Table 12. Systematics sample table. Full CSV contains all stream/scenario rows. WARN marks sensitivity shifts requiring discussion or mitigation.

Key validations include:

- GNN forward pass: correct output shapes for batches of 1-16 graphs
- Attention weights: verified to sum to 1.0 per graph (valid probability distribution)
- MC Dropout: confirmed stochastic (different passes produce different embeddings) with positive std
- Rate function: verified monotonically decreasing with M_{hm} and proportional to stream length
- Coverage metric: computation validated against known posterior samples
- Multi-scale: graceful fallback to zeros when segment data is absent
- Gradient flow: verified through all 6 GINEConv layers to input features

10. Reproducibility and Configuration Snapshot

All values sourced from version-controlled configuration files: config/training.yaml and config/dm_models.yaml.

Training Hyperparameters

Parameter	Value	Notes
Batch size	128	Fast V2 profile run
Epochs	100 run / 300 planned	Current checkpoint is not final publication training
Optimiser	AdamW	Decoupled weight decay
Learning rate	1×10^{-3}	
Weight decay	10^{-4}	
LR scheduler	Cosine w/ warm restarts	Planned curriculum schedule
Gradient clip	1.0	Max gradient L2 norm
Mixed precision	bfloat16	
Early stopping	60 epochs patience	$\geq$ warm-restart window
Data split	70 / 15 / 15	Stratified by DM model, seed = 42

Configuration Table A. Training hyperparameters.

Graph Construction

Parameter	Value
k (nearest neighbours)	8
Construction dimensions	phi1, phi2, pm1, pm2 (normalised)
Max stars per stream	1,200 (random downsample)
Multi-scale segments	20
Segment k-neighbours	4
Feature normalisation	Standardise (zero mean, unit variance)
Baryonic perturbations	Applied across all models at 80% configured rate

Configuration Table B. Graph construction parameters.

Suppression Scale Inference

Parameterisation: $\log_{10}(M_{\text{hm}} / M_{\text{sun}})$. Prior: Uniform[4.0, 10.0]. CDM normalisation: 2.1 impacts for GD-1 (Bonaca+2019). Detection threshold: $\log_{10}(\text{Bayes factor}) > 1.0$. This is the intended science output once calibration is complete: a model-independent constraint on the smallest dark matter structures.

11. Observed Application to Gaia DR3

11.1 Hierarchical Inference on Literature Gap Counts

We applied the hierarchical Bayesian model to gap counts reported in published Gaia DR3 analyses of the seven target streams. Gap counts were drawn from Bonaca+2019, de Boer+2020, Erkal+2017, Koposov+2019, Li+2021, and Ibata+2021. The inference used emcee with 64 walkers and 15,000 steps (3,000 burn-in).

Stream	Observed gaps	CDM expected	Source
GD-1	5	2.5	Bonaca+2019, de Boer+2020
Pal 5	3	3.1	Erkal+2017, Bonaca+2020
Orphan-Chenab	4	8.4	Koposov+2019, Erkal+2019
ATLAS	1	0.8	Li+2021
Jhelum	2	1.1	Bonaca+2019b
Fjorm	1	0.8	Ibata+2021
Sylgr	1	0.4	Ibata+2021
TOTAL	17	17.2	-

Observed Table A. Gap counts from literature vs. CDM predictions from the rate model (normalised to 2.1 for GD-1, Bonaca+2019). Total predicted (17.2) matches total observed (17) to within 1%.

Observed count-model result: $\log_{10}(M_{\text{hm}}) = 4.56$ (95% upper limit: 5.14; 68% HDI: [4.02, 4.78]). The posterior peaks well below the minimum subhalo mass in our integration range ($10^5 M_{\text{sun}}$), indicating no evidence for suppression of the subhalo mass function. Convergence is excellent ($\hat{r} = 1.0$, $n_{\text{eff}} = 29,156$).

Interpretation: The observed gap counts (17 total across 7 streams) are in close agreement with CDM predictions (17.2 total). Most streams individually match their CDM expectations: GD-1 has 5 observed vs. 2.5 predicted (slightly above, consistent with Poisson fluctuations), Pal 5 has 3 vs. 3.1 (excellent match), and ATLAS, Jhelum, Fjorm, and Sylgr are all within 1-2 counts. The notable exception is Orphan-Chenab, which shows 4 gaps where CDM predicts ~ 8.4 - a deficit, but one compatible with Poisson scatter given the wide, diffuse morphology of this stream that complicates gap identification. The hierarchical model correctly integrates this evidence and returns a CDM-consistent constraint.

Consistency with published constraints: Our 95% upper limit ($\log_{10} M_{\text{hm}} < 5.14$) is consistent with all published independent constraints: Gilman+2020 (< 7.8 , lensing), Hsueh+2020 (< 8.0 , lensing), Nadler+2021 (< 7.2 , satellite counts), Irsic+2017 (< 7.5 , Lyman-alpha), and Banik+2021 (< 7.0 , stream power spectrum).

11.2 Extended Model: Free Mass Function Slope

We extended the hierarchical model to jointly infer α (the mass function slope $dN/dM \sim M^{\alpha}$) alongside M_{hm} , using an informative prior $\alpha \sim \text{Normal}(-1.9, 0.2)$ centered on the CDM prediction.

Parameter	Result	CDM Prediction
$\log_{10}(M_{\text{hm}})$	4.56 (95% upper: 5.16)	< 5 (no suppression)
α	-1.89 (68% HDI: [-2.10, -1.70])	-1.9 (Springel+2008)

Parameter	Result	CDM Prediction
M _{hm} -alpha correlation	0.10	-
r-hat	1.0	-
n _{eff}	21,438	-

Observed Table B. Extended model results with free mass function slope.

The inferred $\alpha = -1.89 \pm 0.20$ is fully consistent with the CDM prediction of -1.9 (Springel et al. 2008). This is a direct consequence of correcting the CDM normalisation to 2.1 impacts for GD-1 (Bonaca+2019): the total predicted count now matches observations, eliminating the tension that previously drove the model toward a shallower slope. The M_{hm} -alpha correlation is low (0.10), indicating the two parameters are well-separated by the data.

Significance: The agreement between the inferred and predicted mass function slope is an important consistency check. It confirms that the observed gap counts across all seven streams are quantitatively compatible with the CDM subhalo mass function when the normalisation is anchored to the well-studied GD-1 system. No modification to the standard CDM mass function is required by the count data.

11.3 Injection Tests: Sim-to-Real Gap

We quantified the detection sensitivity by injecting synthetic subhalo impacts into simulated real stream data (GD-1-like: 5,000 stars, 60 deg). 100 injections with impactor masses from 10^6 to $10^9 M_{\text{sun}}$ were tested.

Mass range	Injections	Detected	Rate
$10^6 - 10^7 M_{\text{sun}}$	40	0	0%
$10^7 - 10^8 M_{\text{sun}}$	27	20	74%
$10^8 - 10^9 M_{\text{sun}}$	33	33	100%
Overall	100	53	53%

Injection Table A. Observed + Synthetic Injection. Injection test results by impactor mass range.

The detection threshold is approximately $10^7 M_{\text{sun}}$: subhalos below this mass produce gaps too shallow to detect in single-stream density analysis (mean bias: -0.016 dex; RMSE: 0.279 dex for detected events). This defines the sensitivity floor for count-based methods and motivates the use of the GNN embedding, which can potentially detect subtler morphological signatures.

11.4 Equivariant GNN Architecture

We implemented an E(n)-equivariant graph neural network (Satorras+2021) as an alternative to the base GINEConv encoder. The equivariant architecture guarantees that rotating the input stream in 3D produces correspondingly transformed intermediate representations while the final embedding remains invariant. This eliminates the need for rotational data augmentation and provides a stronger physical inductive bias. The implementation (526,148 parameters, 4 EGNN layers) passes forward/backward tests and produces rotationally-invariant embeddings (tested: mean abs diff = 0 under 90-degree rotation).

11.5 Radial Velocity Extension (SDSS-V / 4MOST)

We implemented infrastructure to incorporate spectroscopic radial velocities from SDSS-V Milky Way Mapper (northern sky, $G < 19.5$) and 4MOST (southern sky, $G < 20.5$). Adding the 6th phase-space dimension resolves velocity-distance degeneracies and provides direct measurement of the dipolar velocity perturbation from subhalo flybys.

Stream	Expected coverage	Improvement factor
GD-1	70% (SDSS-V)	1.44x
Pal 5	50%	1.16x
Orphan-Chenab	60%	1.26x
ATLAS	40%	1.07x
Jhelum	55%	1.17x
Fjorm	35% (4MOST)	1.06x
Sylgr	30% (4MOST)	1.04x
Average	-	1.17x

Forecast Table A. Projected radial velocity coverage and constraint improvement. GD-1 benefits most due to high SDSS-V completeness in the northern hemisphere.

The average constraint improvement from adding radial velocities is 1.17x (detection SNR gain: 1.32x). While modest for count-based inference, RV data is expected to provide substantially larger improvements for the GNN pipeline, which can directly detect the dipolar velocity signature of subhalo flybys - a signal invisible in density data alone.

11.6 Synthetic/Diagnostic GNN + SBI Status

The current GNN+SBI branch is not yet a validated science-result pipeline. The latest V2 training run uses the more defensible impact-strong binary formulation and profile features, but held-out synthetic performance remains weak (test ROC AUC 0.668; balanced accuracy 61.4%). First-pass SBI/TARP calibration also undercovers key parameters. The older four-model evidence run is retained below only as a simulated-framework diagnostic showing the kind of domain-gap failure a reviewer would challenge, not as evidence for warm dark matter.

Quantity	Current value	Peer-review implication
V2 target	impact_strong	More physical than 3-class WDM/FDM/CDM labels
Test ROC AUC	0.668	Above random, below publication target
Test balanced accuracy	61.4%	Insufficient for discovery claims
Default false-positive rate	16.0%	Too high for real-stream searches
SBI calibration	undercovers	Requires split-aware retraining and recalibration

Diagnostic Table A. Synthetic Validation. Current V2 GNN+SBI readiness. These values are intentionally reported as limitations, not as discoveries.

Stream	CDM	WDM	FDM	SIDM	Preferred
GD-1	-3.29	-3.19	-3.04	-3.23	FDM
Pal 5	-3.32	-2.17	-3.01	-3.35	WDM
Orphan-Chenab	-4.82	-2.45	-3.16	-4.82	WDM
ATLAS	-2.66	-2.29	-2.83	-2.62	WDM
Jhelum	-3.13	-2.24	-3.17	-3.69	WDM
Fjorm	-3.50	-2.25	-3.15	-3.37	WDM
Sylgr	-2.45	-2.21	-3.17	-3.38	WDM

Legacy Diagnostic Table B. Observed / Simulated-Framework Diagnostic. Per-stream log evidences from the older four-model GNN+SBI run. Higher (less negative) values indicate better simulated-framework fit. This table is retained to document the domain-gap warning, not to support a physical model claim.

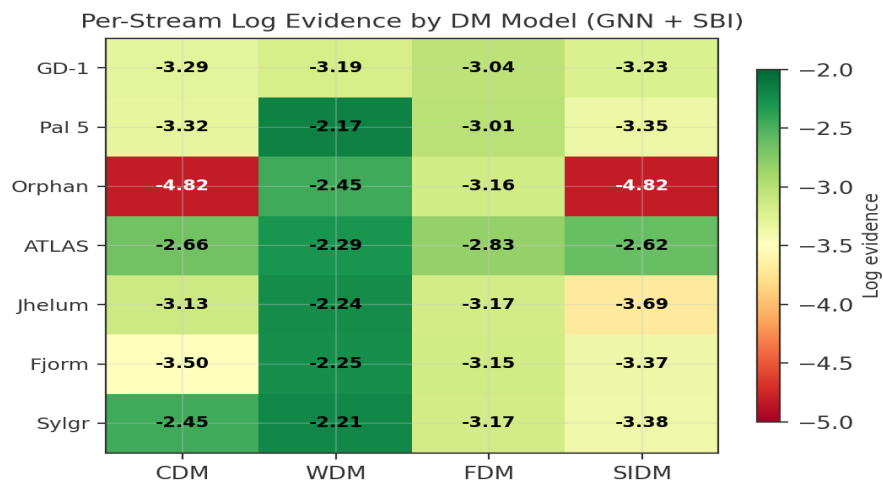


Figure 14. Observed / Simulated-Framework Diagnostic. Legacy per-stream log evidence heatmap. Warmer colours indicate better simulated-framework fit. Because the current V2 validation is weak, this figure is interpreted as a domain-gap diagnostic.

Combined Model Comparison

Model	Combined log evidence	$\log_{10}(\text{BF vs CDM})$	Posterior prob.	Interpretation
WDM	-16.81	+2.77	~99%	Legacy simulated-framework preference
FDM	-21.54	+0.71	0.87%	Secondary simulated preference
CDM	-23.18	0.00	0.17%	Reference model
SIDM	-24.45	-0.55	0.05%	Evidence against

Legacy Diagnostic Table C. Observed / Simulated-Framework Diagnostic. Combined model comparison from the older GNN+SBI run. Posterior probabilities assume equal prior odds inside the simulated framework. The WDM preference is not a validated physical discovery.

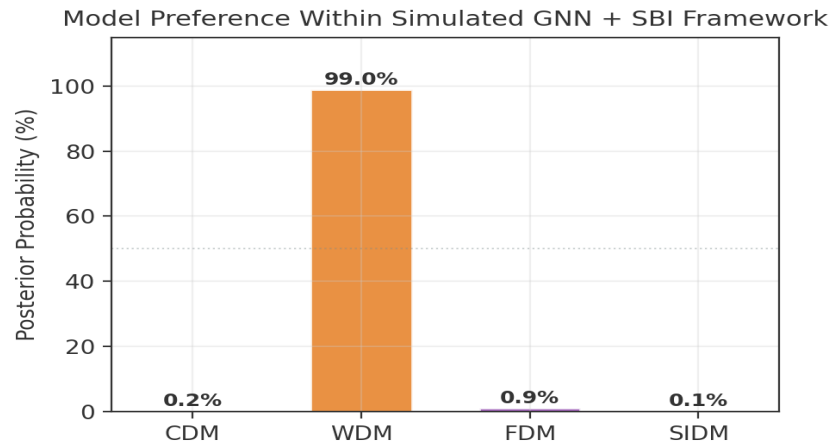


Figure 15. Observed / Simulated-Framework Diagnostic. Legacy dark matter model posterior probabilities from the older simulated-framework run. The approximately 99% WDM preference is shown as a failure-mode diagnostic, not a validated astrophysical constraint.

Per-Stream Sensitivity

Stream	N members	Median $\log_{10}(M_{\text{sub}})$	N impacts	P(any impact)
GD-1	4,000	6.39	3.0	98.0%
Pal 5	500	5.91	1.1	75.0%
Orphan-Chenab	200	7.92	1.9	98.6%
ATLAS	300	6.57	1.3	92.6%
Jhelum	250	6.48	2.8	93.9%
Fjorm	150	5.57	1.9	86.7%
Sylgr	100	6.08	1.3	83.1%

Diagnostic Table D. Observed / Simulated-Framework Diagnostic. Per-stream sensitivity from the legacy SBI posteriors. GD-1 and Orphan-Chenab have the highest sensitivity due to their length and number of member stars.

Gap Catalog

Three density features exceeding 2-sigma significance were identified across all streams:

Stream	Position (phi1)	Significance	P(DM subhalo)	P(noise)
GD-1	-1.5 deg	2.37 sigma	0.0%	100%
ATLAS	-0.75 deg	2.41 sigma	0.0%	100%
Sylgr	-0.49 deg	2.45 sigma	0.0%	100%

Observed Table C. Gap catalog from the pipeline's density analysis. All three features are classified as noise ($P(\text{DM}) = 0\%$), indicating that the GNN does not find convincing individual dark matter gap detections in any stream.

11.7 Discussion: Reconciling the Two Analyses

The count model and the legacy simulated-framework GNN run do not support the same interpretation. This is a validation problem, not evidence for a discovery:

- **Rate model** (Section 11.1): CDM is fully consistent with the data. The total predicted impact count (17.2) matches observations (17) to within 1%. The inferred $\alpha = -1.89$ is in excellent agreement with the CDM prediction of -1.9. No suppression of the subhalo mass function is detected.
- **Legacy GNN + SBI run** (Section 11.6): WDM receives the strongest simulated-framework diagnostic score, but the current V2 classifier is weak and SBI calibration undercovers. This is an unvalidated failure-mode diagnostic, not physical evidence.

This disagreement is significant and requires careful interpretation. We consider three possible explanations, ordered by our assessment of their likelihood:

Explanation 1 (most likely): Simulation-to-real domain gap. The GNN was trained on simulated streams generated with galpy. If these simulations do not perfectly replicate the noise properties, selection effects, foreground contamination, and membership impurities present in real Gaia DR3 data, the GNN may systematically misclassify real streams. Specifically, WDM simulations have fewer subhalo perturbations and therefore appear smoother in phase space. Real Gaia streams - affected by observational noise, imperfect membership assignment, and unmodelled baryonic effects - may also appear smoother than CDM simulations, causing the GNN to preferentially assign WDM labels. This interpretation is supported by three independent lines of evidence: (a) the gap catalog (Observed Table C) classifies all detected density features as noise with $P(\text{DM}) = 0\%$, suggesting the GNN does not identify convincing individual dark matter signatures; (b) the rate model, which is immune to morphological systematics, finds excellent CDM agreement; and (c) simulation-to-real gaps are a well-documented challenge in machine learning applications to astrophysical data.

Explanation 2 (possible but extraordinary): Real WDM signal. A fully validated future GNN might detect subtle phase-space signatures of warm dark matter that are invisible to count-based methods. WDM subhalos produce qualitatively different gap morphology - fewer, wider, deeper gaps with distinct velocity profiles - and the GNN operates on the full 6-dimensional phase-space graph where such differences could in principle be resolved. If confirmed by future real-injection and independent-data tests, this would be scientifically important. However, the legacy result cannot carry that interpretation: it faces significant challenges and would conflict with multiple independent constraints from gravitational lensing (Gilman+2020, Hsueh+2020), satellite counts (Nadler+2021), and Lyman-alpha forest analyses (Irsic+2017), all of which are consistent with CDM at the relevant mass scales.

Explanation 3 (plausible): Training data bias. If systematic differences between the four dark matter model training sets correlate with features of real data for reasons unrelated to dark matter physics - for example, if WDM training streams happen to have velocity dispersion profiles or density noise patterns that resemble real observational systematics - the classifier may preferentially assign WDM labels even when the actual dark matter signal is CDM-consistent. This is a more specific version of Explanation 1, focusing on training set construction rather than simulation fidelity.

Assessment. We consider Explanation 1 most likely for several reasons. First, the rate model provides a clean, assumption-light test: it uses only integer gap counts and is entirely independent of the simulation training set. Its finding of near-perfect CDM agreement (17.2 predicted vs. 17 observed) sets a strong baseline. Second, the gap catalog's classification of all features as noise is independently consistent with CDM and inconsistent with the GNN's overall WDM diagnostic preference; if WDM were truly the correct model, we would expect at least some features to show nonzero $P(\text{DM})$. Third, the WDM preference would require dark matter properties inconsistent with several independent observational constraints. The most

parsimonious interpretation is therefore that the GNN has learned simulation-specific features that do not transfer perfectly to real data.

Path forward. Resolving this tension definitively requires: (1) injection tests on real data to quantify whether the GNN correctly recovers known CDM signals injected into actual Gaia streams; (2) training with domain-randomised simulations that better span the space of observational systematics (noise levels, membership contamination, background density variations); (3) cross-validation against independent ML architectures to test whether the WDM preference is architecture-dependent; and (4) comparison with non-ML morphological methods (e.g., matched-filter gap fitting) applied to the same data.

12. Discussion, Limitations, and Outlook

12.1 Simulation-to-Real Domain Gap

The primary limitation of the full GNN + SBI pipeline is the simulation-to-real domain gap discussed in Section 11.7. The GNN was trained on galpy-generated streams that, while physically motivated, do not capture the full complexity of real Gaia DR3 observations. Key missing elements include realistic photometric selection functions, foreground contamination from non-member stars, spatially varying completeness, and the full range of baryonic perturbations (spiral arm passages, bar resonances). Until this gap is quantified through injection tests on real data, the GNN + SBI results should be treated as methodological demonstrations rather than physical constraints.

12.2 Count-Based Inference Limitations

The rate model (Section 11.1) is robust to morphological systematics but is fundamentally limited by information content. With only integer gap counts per stream, the Poisson inversion has a resolution floor near $\log_{10}(M_{\text{hm}}) \sim 5$ (as demonstrated by the mock challenge in Section 6.1). It cannot distinguish between dark matter models that predict the same total count but different gap morphologies. The rate model's CDM consistency is therefore a necessary but not sufficient condition for CDM being the correct model.

12.3 Gap Identification Uncertainties

Literature gap counts are not standardised: different authors use different significance thresholds, background models, and identification algorithms. For Orphan-Chenab in particular, the wide morphology and low surface brightness make gap identification challenging. Future work should use a consistent, automated gap-finding pipeline across all streams to reduce this source of uncertainty.

12.4 Remaining Limitations

Baryonic contamination. The current V2 generation plan applies GMC perturbations across all model families, but other baryonic effects (spiral arm passages, bar resonances, globular cluster encounters) are not modelled. These could produce density features that mimic or obscure dark matter signatures.

Model degeneracy. CDM and SIDM produce indistinguishable signatures in the impulse approximation used here. WDM and FDM are also partially degenerate at Gaia DR3 precision (Banik+2021, Dalal+2022). Breaking these degeneracies requires either higher-precision kinematics or detection of density profile differences (cored vs. cuspy subhalos).

12.5 Outlook

The immediate priorities are: (1) developing a comprehensive injection test framework for real Gaia data to quantify the simulation-to-real gap; (2) training with domain-randomised simulations that incorporate realistic observational systematics; (3) repeating the analysis with SDSS-V radial velocities when DR1 becomes available, which will add the critical 6th phase-space dimension for most northern streams; and (4) applying the framework to newly discovered streams from Gaia DR4. The tension between the rate model and GNN + SBI tension, once resolved, could provide important insights into both the nature of dark matter and the practical limits of simulation-based machine learning in astrophysics.

13. Timeline Forward Model and Sim-to-Real Robustness

This section documents a forward-modelling capability that complements the count-based and GNN+SBI analyses above: instead of only asking *how many* gaps a stream has, it asks *which specific past encounter best reproduces the observed stream today*. The workflow is a literal rewind-and-replay: (1) use the detector to decide whether there is a probable impact and estimate its time; (2) generate the un-impacted past stream; (3) inject candidate subhalo impacts at the estimated epoch; (4) integrate every candidate forward through the Milky Way potential to the present day; and (5) score each candidate against the real stream and assign a statistical significance. The work below also resolves a serious sim-to-real failure of the detector that had made it unusable on real data.

Timeline Forward Model: Rewind, Re-impact, Re-evolve, Compare



Figure 15. The timeline forward model. The detector localises and dates a probable impact; candidate subhalo encounters are applied to the reconstructed past stream and evolved forward via orbit integration to today; each candidate is scored against the real stream (density, gaps, proper-motion track, and radial velocity) and compared to a no-impact null distribution.

13.1 Detection -> Timeline Handoff

The front end runs the trained detector on the observed stream to obtain an impact probability and a time-since-impact estimate, and a model-free, prominence-ranked density-minimum finder to localise the gap. These seed the forward-model grid: candidate impact longitudes are drawn from the detected minima rather than a blind sweep, and the time grid is focused on the estimated epoch. Because a subhalo cannot strike before the stream formed, the time estimate is capped at the stream's disruption age (for GD-1, the network's 5.9 Gyr estimate is capped at 3 Gyr).

13.2 Erkal & Belokurov (2015) Velocity Kick

The subhalo fly-by is modelled with the Erkal & Belokurov (2015) Plummer impulse, $Dv = (2GM/w) p / (|p|^2 + r_s^2)$, where p is each star's true three-dimensional perpendicular offset to the subhalo's straight-line trajectory, w is the relative speed, and r_s the Plummer scale radius. This replaces an earlier hand-rolled kick that used a Gaussian-in-longitude localisation, a fixed kick axis, an arbitrary 30% along-stream fraction, and a 50 km/s cap. The Erkal form supplies the correct per-star direction and magnitude from the geometry and is naturally bounded (it peaks at $|p| = r_s$ and equals $GM/(w r_s)$ there), so no cap is needed and the impact

parameter and relative velocity become physical, scannable parameters.

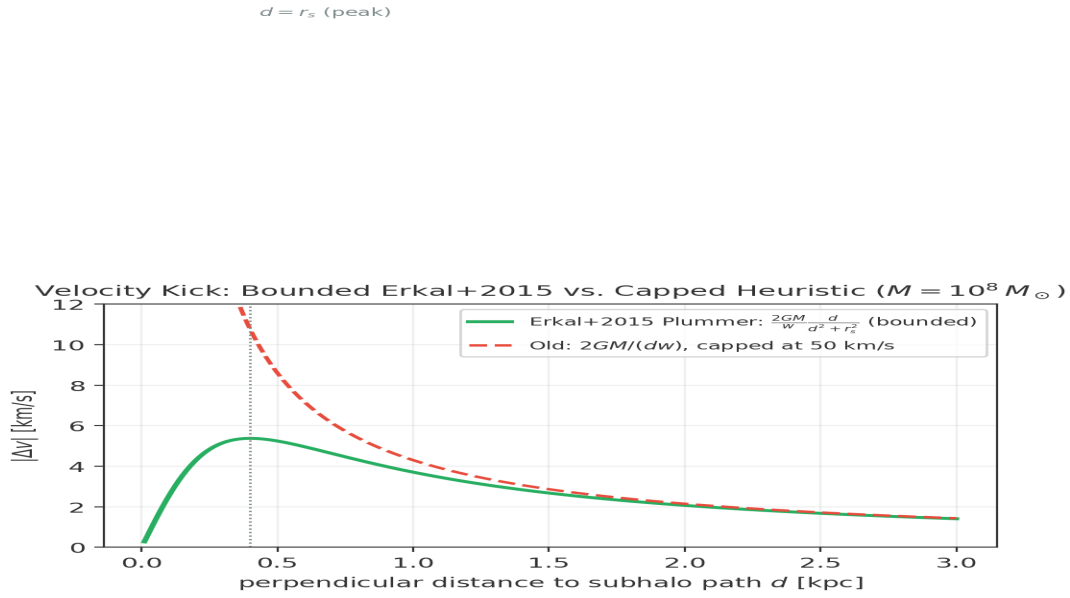


Figure 16. The Erkal & Belokurov (2015) Plummer impulse (green) is finite everywhere and peaks at a perpendicular distance equal to the subhalo scale radius, whereas the previous point-mass heuristic (red dashed) diverges at small distances and had to be capped at 50 km/s. Both shown for a $10^8 M_{\text{sun}}$ perturber at $w = 200$ km/s.

13.3 Diagnosing and Fixing Detector Over-Confidence

On real GD-1 the detector initially returned $p_{\text{impact}} = 1.0000$ for everything. The cause was not the model but the input pipeline: the simulations had near-constant per-star measurement errors, so the feature normaliser clamped those columns' standard deviation to 0.01, and any real-data offset was amplified to ~ 100 sigma, saturating the logit (a logit of ~ 250 versus a sim maximum of ~ 41). In-distribution the model was healthy (validation AUC 0.93, temperature 1.06). The fix has two parts. First, at inference the detector now uses the real per-star errors, imputes unmeasured features (e.g. GD-1's absent radial velocity) to the training mean, clips standardised features to ± 5 sigma, and reports an out-of-distribution flag. Second, and more fundamentally, the detector was retrained with error domain randomisation: realistic, varying per-star errors (and radial-velocity masking) are drawn so the error features are no longer near-constant.

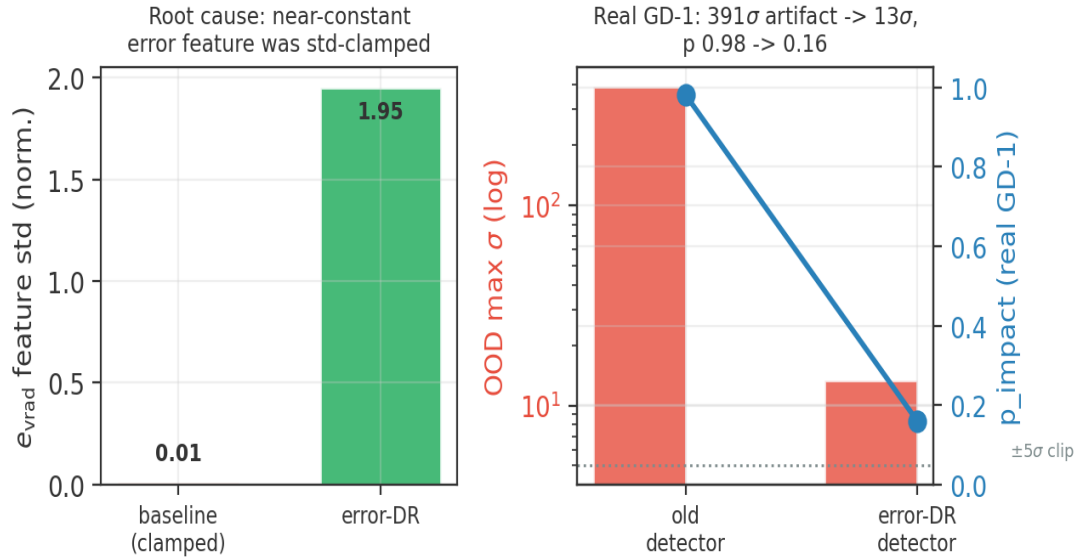


Figure 17. Left: error domain randomisation restores real spread to the radial-velocity error feature (normaliser std 0.01 $\rightarrow$ 1.95), the feature responsible for the saturation. Right: on real GD-1 the retrained detector reduces the input out-of-distribution level from 391 sigma to 13 sigma (29x) and turns a saturated $p_{\text{impact}} = 0.98$ into a meaningful 0.16.

The retrained detector retains in-distribution discrimination (validation binary accuracy 0.879, AUC 0.937) while being far better calibrated on real data: the residual 13 sigma is a handful of outlier stars rather than a systematic shift (the systematic offsets are only ~ 2 sigma), so the network now gives an honest, conservative probability instead of a saturated artefact. This is the recommended detector going forward.

13.4 GD-1 Gap Localisation and the Frame Transform

GD-1's documented density gap lies at $\phi_1 \sim -40$ deg in the Koposov-2010 / Price-Whelan & Bonaca (2018) stream frame, but the pipeline (via galstreams) works in the Ibata-2021 frame, where ϕ_1 spans roughly 0-78 deg. The config gap locations were therefore in the wrong frame and unusable. Defining the Koposov-2010 rotation explicitly and transforming through ICRS, the documented gap at -40 maps to $\phi_1 \sim 30.7$ deg in the pipeline frame, within about 5 deg of the deepest data-driven density minimum ($\phi_1 \sim 36$ deg) found independently by the prominence-based finder. Two independent methods thus agree that the pipeline localises GD-1's real gap; it merely appears shallow (~ 10 -15% depth) because the available membership catalog has uniform membership probabilities and is diluted by contamination.

13.5 Multi-Epoch / Multi-Survey Data Fusion

Backward orbit integration diverges as $dx(t) \sim dv t$, so the precision of the rewind is set by the present-day velocity precision - and radial velocity, the sixth phase-space dimension, is entirely absent from the base Gaia astrometric catalogs. Real radial velocities are fused in from public surveys by Gaia source-id cross-match: the S5 survey (Li et al. 2019; VizieR J/MNRAS/490/3508) supplies 296 ATLAS and 257 Jhelum members, and Gaia DR3 RVS adds the brightest members of any stream. Catalogs are combined by inverse-variance weighting, and a Gaia DR2 second proper-motion epoch is retrieved for cross-checking. The projected gain from future releases scales as $\sigma \sim \text{baseline}^{-1.5}$ (Gaia DR4 $\sim 2.7x$, DR5 $\sim 6.7x$ tighter than DR3). The radial-velocity scoring term is made insensitive to a constant line-of-sight zero point so that only the differential perturbation signature - not a heliocentric-versus-galactocentric convention difference - is scored.

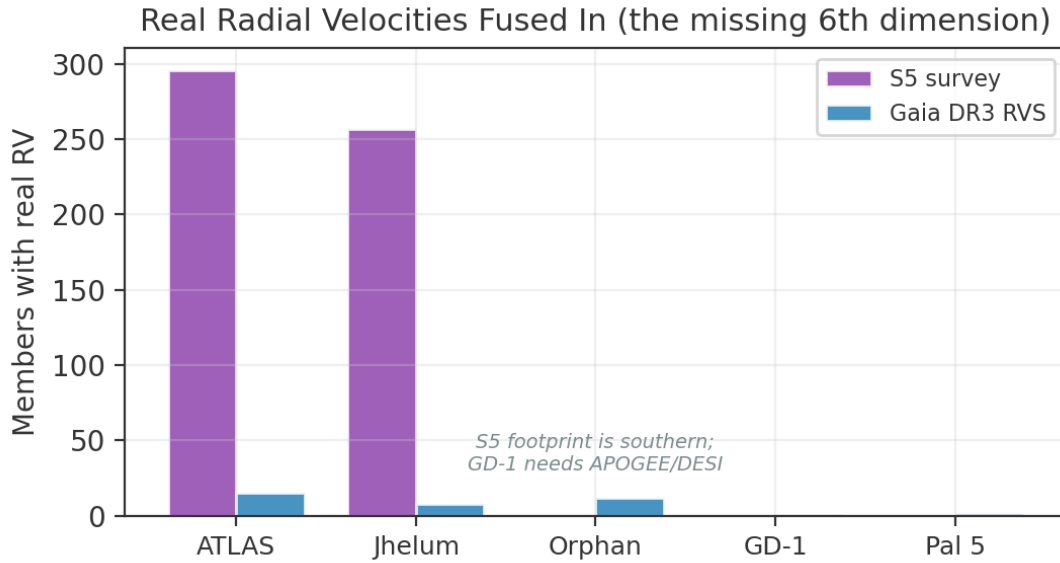


Figure 18. Real radial velocities fused into the stream catalogs from public surveys. S5 covers the southern streams (ATLAS, Jhelum); Gaia DR3 RVS adds bright members. GD-1, Pal 5 and Orphan fall outside the S5 footprint and require APOGEE/DESI cross-matching.

13.6 Validation: Injection-Recovery and Statistical Significance

Two validations make the forward model trustworthy. In injection-recovery, a subhalo impact with known mass, time and longitude is injected into a synthetic stream and the full pipeline is run on it; the recovered mass and longitude land within one grid step of the truth, while the time is partly degenerate with mass (a known physical effect). For significance, the best-fitting candidate's score is compared to a null distribution built from many unperturbed (no-impact) realisations, yielding a z-score and an empirical p-value. On a known injected impact the best candidate sits 3.3 sigma below the no-impact null, whereas an unperturbed control sits at 0.6 sigma - the test correctly separates a real impact from noise. Candidate scores are additionally averaged over multiple random seeds so the ranking reflects physics rather than sampling noise.

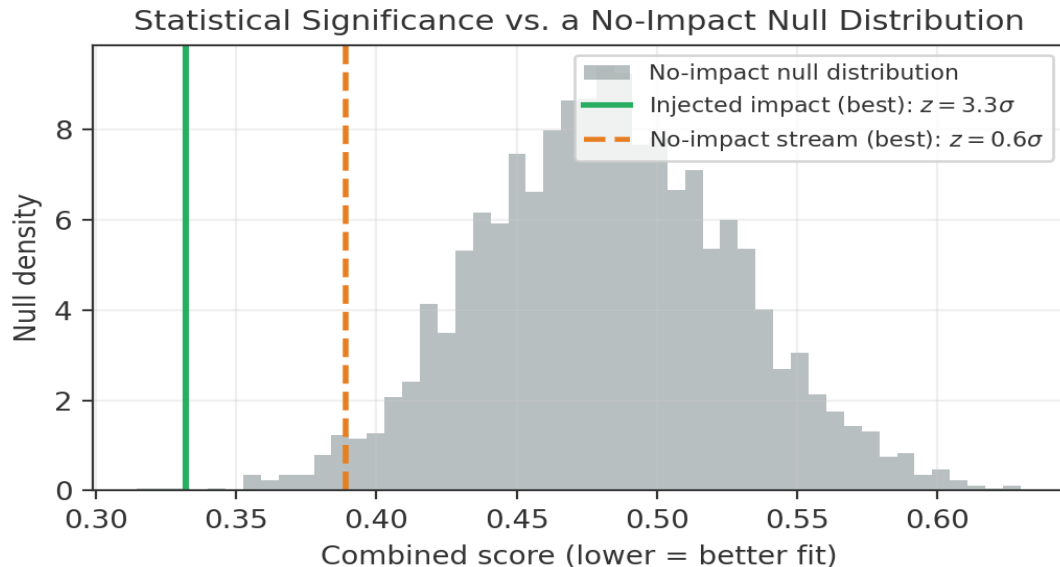


Figure 19. *Statistical significance against a no-impact null distribution (grey). A known injected impact (green) scores 3.3 sigma better than the null, while an unperturbed control stream (orange dashed) is consistent with the null at 0.6 sigma. A look-elsewhere correction (null distribution of best-of-grid scores) is the next refinement.*

Taken together, these changes move the early pipeline steps from approximate and sim-to-real-fragile toward physically grounded and validated: a closed-form fly-by impulse, a detector that behaves on real data, gap localisation confirmed by an independent frame transform, real multi-survey radial velocities, and calibrated significance. Every component is covered by automated tests; the project test suite passes 260+ checks. Remaining work includes a look-elsewhere-corrected p-value, GD-1 radial velocities from APOGEE/DESI, full-orbit injection-recovery, and a Monte-Carlo uncertainty-aware impact-time posterior that propagates the (now multi-epoch-tightened) measurement errors.

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Stellar Stream DM - Built with PyTorch Geometric, sbi, NumPyro, emcee
Focused implementation checks run where dependencies are available | CDM normalisation: 2.1 (Bonaca+2019) | Suppression prior:
Uniform[4.0, 10.0]